1 Background

Discuss the importance of magnetic flux distribution and how it affects the performance of these devices. Highlight the challenges associated with achieving uniform magnetic flux distribution and the potential benefits of using multipermeability cores.

Soft magnetic cores are a cruical part in many inductors and transformers, guiding magnetic flux and influence their electromagnetic performance [1]. Magnetic components typically take up a notable portion of size and weight of a power converter. More compact designs of magnetic components greatly improve power density of the system, although are constrained by fabrication and design constraints [2]. Therefore, there is ongoing research in optimizing The performance of these devices is heavily influenced by the distribution of magnetic flux within the core material. A nonuniform magnetic flux distribution lead to higher core losses [3] and lower localized saturation [4]. Having uniform distribution of magnetic flux throughout an inductor core, leads to the reduction in volume of the inductor without sacrificing its inductance [5]. Furthermore, decreasing the core volume leads to lower winding loss due to reduced winding lenght. Therefore, uniform magnetic flux distribution is desired.

One way of achieving uniform magnetic flux distribution is

It has been shown that 3-10 multipermeability layers are optimal, as larger amount exponentially comes closer to continuous permeability [6].

Why do magnetic cores need multipermeability? Establish the engineering problem. - How does a nonuniform flux distribution constrain core size, inductance, saturation and power density? - How does spatially varying permeability redistribute magnetic flux? - Under what operating conditions does a multipermeability core outperform a uniform-permeability core?

How have researchers manufactured multipermeability cores? Establish the state of the art and its limitations. - How did Liu et al. create their different permeability regions? - What other methods have been used: separate magnetic materials, geometric changes, composition gradients or process-induced material changes? - Are existing approaches limited to discrete regions, or have continuous gradients been demonstrated? - What manufacturing complexity, material compatibility or geometric constraints do these approaches introduce? - What remains difficult to manufacture, even if the electromagnetic design is understood?

Why pellet extrusion for magnetic composites? Justify your manufacturing method rather than assuming it is better. - What magnetic composites have already been processed using pellet-fed material extrusion? - Why use pellets instead of filament: feedstock preparation, achievable particle loading, material cost or processing flexibility? - Can the magnetic composite be diluted with pellets of its unfilled polymer matrix? - How do particle loading, viscosity, nozzle size and printability constrain the material? - What advantages does pellet extrusion offer over existing filament-based or paste-extrusion approaches for this application?

Can pellet extrusion produce a controlled permeability gradient? This is the key technical bridge in your argument. - Has anyone independently metered magnetic-composite and unfilled pellets during a single print? - Can their mixing ratio be varied continuously, and how quickly does the extrudate respond to a change in feed ratio? - Does the screw provide sufficient distributive mixing, or are additional mixing features required? - How uniform is the magnetic-particle distribution within the extrudate? - How does pellet feed ratio translate into actual particle concentration and effective permeability? - What are the minimum achievable gradient length, transition width and repeatability?

Why vary composition instead of geometry? Your most important justification question. - Can a continuous composition gradient achieve a magnetic response that is difficult to approximate using discrete geometric regions? - Is the desired gradient finer than the spatial resolution of the printing process? - Can composition grading avoid adding multiple magnetic materials or

complex internal geometries? - Does changing composition provide a useful degree of freedom beyond geometry, or can geometry achieve the same performance with less complexity? - Under equal volume, material and operating constraints, which approach produces the best core performance?

What is actually missing from the literature? Define a defensible research gap. - Has pellet extrusion already been used to produce spatially graded magnetic composites? - Has the local permeability of such a printed gradient been measured? - Has a graded pellet-extruded core been tested as an inductor? - Has its performance been compared against a uniform core and a geometry-optimised core? - If these demonstrations are missing, is the missing contribution the manufacturing process, the magnetic characterisation, the core design or the device-level performance?

2 Problem Statement

2.1 Research Question

To address the identified problem, the research question is defined as follows:

2.2 Requirements

To address the research question and capture the overarching scientific plan, the following requirements have been defined:

2.3 Desirables

Beyond the essential requirements, additional desirable outcomes are below:

2.4 Delimitations

The project will employ commercially available additive manufacturing equipment, modified for the task at hand.

3 Methodology

3.1 Requirements

To address this research question, the following requirements have to be met:

3.2 Desirables

Beyond the essential requirements, additional desirable outcomes are below:

References

[1] Vencislav Cekov Valchev and Alex Van van den Bossche. *Inductors and Transformers for Power Electronics*. CRC Press, 2018.

[2] Charles R. Sullivan, Di Yao, Garet Gamache, Alexander Latham, and Jizheng Qiu. Passive Component Technologies for Advanced Power Conversion Enabled by Wide-Band-Gap Power Devices. 41(8):315–330.

[3] Edris Agheb and Hans Kristian Høidalen. Modification of empirical core loss calculation methods including flux distribution. 7(5):381–390.

[4] Kazuhiro Umetani. Improvement of saturation property of iron powder core by flux homogenizing structure. 8(6):640–648.

[5] Lanbing Liu, Chao Ding, Shengchang Lu, Ting Ge, Yi Yan, Yunhui Mei, Khai D. T. Ngo, and Guo-Quan Lu. Design and Additive Manufacturing of Multipermeability Magnetic Cores. 54(4):3541–3547.

[6] Laili Wang, Zhiyuan Hu, Yan-Fei Liu, Yunqing Pei, and Xu Yang. Multipermeability Inductors for Increasing the Inductance and Improving the Efficiency of High-Frequency DC/DC Converters. 28(9):4402–4413.